WORKSHEET

Chapter 9: Solids of revolution

1 Use suitable definite integrals to find the volumes (in cubic units) that are formed by rotating the following curves about the x-axis between the limits shown.

a $y = x^2 + 1$ from $x = 0$ to $x = 1$.

b $y = \sqrt[5]{x}$ from $x = 0$ to $x = 1$.

c $y = x^3$ from $x = 1$ to $x = 2$.

d $y = e^{-x}$ from $x = -3$ to $x = -1$.

e $y = 1 - \cos(x)$ from $x = 0$ to $x = 2\pi$.

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2 Use suitable definite integrals to find the volumes (in cubic units) that are formed by rotating the following curves about the y-axis between the limits shown.

a $y = x^2$ from $x = 0$ to $x = 1$.

b $x = y^3$ from $y = 0$ to $y = 1$.

c $y = \sqrt[7]{x^5}$ from $y = 0$ to $y = 1$.

3 A semicircle of radius r has the equation $y = \sqrt{r^2 - x^2}$. Calculate the volume of the solid formed when the semicircle is rotated about the x-axis. Comment on the result.

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4 Calculate the resulting volume generated when each of the following relations is rotated about the Cartesian axis indicated between the limits shown.

a $y = x^2 - 1$ about the x-axis from $x = -1$ to $x = 1$.

b $y = \frac{1}{x}$ about the y-axis from $y = 1$ to $y = 2$.

c $y = (2 + x)^2$ about the x-axis from $x = 0$ to $x = 1$.

d $y = \cos(3x)$ about the y-axis from $x = 0$ to $x = \frac{\pi}{6}$.

e $x^2 - y^2 = 9$ about the x-axis from $x = -4$ to $x = -3$.

f $y = x^2$ about the y-axis from $y = 0$ to $y = 2$.

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5 The equation of an ellipse is given by $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a and b are the radii at the x- and y-axes respectively. Calculate the volume of the solid generated when:

a the ellipse is rotated about the x-axis

b the ellipse is rotated about the y-axis.

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Answers

All answers are given in cubic units.

1 **a** $\frac{28\pi}{15}$

b $\frac{5\pi}{7}$

c $\frac{127\pi}{7}$

d $-0.5\pi e^2(e^4 + 1)$

e $3\pi^2$

2 **a** $\frac{\pi}{2}$

b $\frac{\pi}{7}$

c $\frac{5\pi}{19}$

3 $V = \frac{4\pi r^3}{3}$. This represents the volume of a sphere of radius r.

4 **a** $\frac{16\pi}{15}$

b $\frac{\pi}{2}$

c $\frac{211\pi}{5}$

d $\frac{\pi(\pi - 2)}{9}$

e $\frac{10\pi}{3}$

f 2π

5 **a** $\frac{4\pi ab^2}{3}$

b $\frac{4\pi a^2 b}{3}$

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